

Week 7: Deliverable 2- AI-Generated KPI Summary

Section 1: Board-level KPI Summary

StaySense BI – Executive KPI Summary

The StaySense BI dashboard, developed using the NYC Airbnb 2019 dataset, provides valuable insights into listing performance across different neighbourhood groups and room types. Analysis indicates that Manhattan consistently records the highest average listing prices, reflecting strong demand for premium accommodations, while Brooklyn offers a more balanced combination of affordability and listing volume.

Customer engagement, measured through the number of reviews, is highest for Entire Home/Apartment listings, suggesting that travellers generally prefer complete-property accommodations. Private Rooms represent the largest share of available listings, highlighting their importance in the overall Airbnb marketplace. Price distribution analysis also reveals significant variation across listings, indicating the need for differentiated pricing strategies rather than a uniform pricing approach.

The three primary KPIs monitored in this dashboard are Average Listing Price, Number of Customer Reviews, and Room Type Distribution. Together, these metrics provide stakeholders with a comprehensive understanding of market trends, customer behaviour, and inventory composition.

Based on these findings, two strategic recommendations are proposed. First, hosts should adopt borough-specific dynamic pricing strategies to better align listing prices with local demand and market conditions. Second, marketing efforts should prioritize Entire Home listings, as they consistently demonstrate stronger customer engagement and greater revenue potential. Continuous monitoring of these KPIs through an interactive BI dashboard will support informed decision-making, improve competitiveness, and help maximize long-term business performance.

Section 2: Changes Log

AI Generated Statement	Final Edited Version	Reason for Change
"Listings are performing well across all boroughs."	Manhattan records the highest average listing prices, while Brooklyn offers a balanced market.	Replaced a vague statement with dataset-supported findings.
"Revenue is steadily increasing."	Removed.	The dataset does not contain revenue information.
"Customer satisfaction is high."	Customer engagement is measured through review activity.	The dataset includes reviews but not customer satisfaction metrics.
"Luxury listings dominate the market."	Premium-priced listings are concentrated primarily in Manhattan.	More accurate and supported by the data.
"Marketing should be improved."	Prioritize marketing efforts for Entire Home listings.	Converted a generic recommendation into a specific, evidence-based action.